A small JEPA world model on a low-cost arm: the encoder reads the outcome, the predictor adds nothing to it

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Abstract

A small JEPA world model trained on a low-cost arm reads how an episode ended, and its predictor adds nothing to that reading. We trained the model in the LeWorldModel recipe on teleoperated demonstrations of one tabletop task on an SO-101 arm, then re-tested on it every use of such a model we found in recent papers, each against a bar written into a public register. Every use that needs the predictor to roll forward misses its bar, chunk ranking, surprise monitoring and risk prediction among them. The encoder does better. On policy rollouts it never saw, a linear probe on its last-frame latent separates success from failure about as well as a hand-built colour judge (AUROC 0.95 against 0.94), and better than a frozen DINOv2 encoder or the commanded joints. At the grasp, the imagined latent reads the outcome worse than the current one (0.70 against 0.82). So before building on the predictor of a model this size, probe its encoder. The controls are exploratory, and everything comes from one rig, one model and one data scale.

1 Introduction

A world model is usually valued for its predictor. In the sense of LeWorldModel (LeWM) [1] it has two parts: an encoder that maps camera frames to a latent vector, and a predictor that takes the current latent and a sequence of actions and imagines the latents that follow, both trained together from pixels with a joint-embedding predictive (JEPA) loss. The uses proposed for it lean on the predictor: ranking a policy's candidate action chunks by their imagined outcome, flagging a failure when the observed latent drifts from the predicted one, reading the outcome of a grasp before it happens, planning over several time scales. Each paper reports the use that worked, on its own robot and data.

We had one such model, trained for $3.30 of rented GPU time on our own arm, and wanted to know which of these uses we could build on. So we ran every use we could find on that one model. Each re-test is a row of a register with a bar, a dataset, a result and a verdict, and where we managed it, the bar and the decision rule went into the repository before the result existed (that commit order is all we mean by pre-registration). Checked against the commit history, all 20 rows match a pre-registration record. In 19 the bar was committed before the result, and in 1 (the two-level hierarchical progress use) in the same commit. One row is not a use of our model: it times the pretrained RoboJEPA and V-JEPA 2.1 encoders on our on-board computer [2, 3]. Of the 20 rows, 3 pass and 1 is met as predicted. Another 2 are not proven, and the remaining 14 fail or are no-go.

The first draft of this paper argued something else. Our reading of the failures was that the latent mostly encodes the arm's pose, so a frozen image encoder or the commanded joints would do at least as well in every one of them. We ran those controls on 69 policy rollouts before writing it up, and they did not bear it out: the encoder reads how the episode went better than either control. We rewrote the claim. At this data scale the predictor, not the encoder, is the weak link.

One moment runs through the rest of the paper: the grasp anchor, 15 frames before the gripper first closes. There the current latent reads the outcome at 0.82 and the imagined latent at 0.70, and that gap is the argument in one row.

The controls came after the data, which limits what they can show. Each readout is a linear probe fit with leave-one-session-out (LOSO) cross-validation on the 69 rollouts (44 success, 25 failure, 13 sessions), scored against labels the author gave at rest. The model was trained on teleoperation only and never saw these rollouts. No bar was committed for these comparisons, so everything this paper says about the split between encoder and predictor is exploratory. Of the 3 register passes, 3 have their bar committed before the result, though the register marks the outcome-readout verdict as retrospective.

*Code, data pointers and every number's source: https://github.com/ravediamond/paper-jepa-world-model.

2 Setup

The robot is one SO-101 arm [4] driven through LeRobot [5], with a top and a wrist camera. The task is to put a bowl into another bowl. Our model is trained on `ravediamond/so101_bowls_all_trimmed`, split by grid layout into 157 training and 25 validation episodes. The evaluation rollouts come from a separate dataset, `ravediamond/so101-bowls-policy-rollouts`: SmolVLA [6] runs recorded after our model was trained.

Our model follows the LeWM recipe [1], with a multi-view ViT encoder, an action-conditioned predictor and the SIGReg regulariser. We set it beside four signals, each chosen because it is cheaper than a world model. The frozen DINOv2-S control is a frozen DINOv2 small encoder [7] mean-pooled over the top and wrist views (the design of DINO-WM [8]). The joints control is the commanded joint vector. The colour judge is a hand-built rule on the final frame with no learning, and policy identity is the checkpoint that ran the episode.

In October, a timing test on the on-board computer loaded our checkpoint with `strict=False`. Under the transformers version on that image (5.5.4), 192 encoder tensors did not match and were silently dropped, so the test timed a random encoder. We voided its latents and traced how every other use loaded the model: 26 ran the trained model, 1 ran a random one (that test) and none is unknown. The world-model block of VLA-JEPA, which one diagnostic below uses, loads all 351 of its keys.

3 The register

Table 1 lists the 20 uses and what each takes from our model: the encoder alone (7), the predictor (8) or both (5). Read that column against the verdicts.

Among the encoder uses, reading the outcome works and reading progress does not. The outcome readout at rest passes, and at the grasp anchor the current latent reads the grasp at 0.82. The four uses that ask the latent for progress or task stage all fail: progress monotonicity, two-level hierarchical progress [10], the latent stage tracker and the latent-distance scale of SURVEYOR [13]. The pose probes of LeJEPA [12] say what else the latent holds. A linear readout recovers about half of the joint variance from our encoder (R^2 0.565), less than from the frozen DINOv2-S control (0.867).

The predictor does predict its own latent: at K = 30 its open-loop error is 0.31, against 0.89 for copying the last latent. What fails is every use of that prediction. Chunk ranking [1] puts the executed chunk first in 0.60 of anchors against a bar of 0.8, while the commanded actions alone reach 0.68–0.90 (so the cheapest signal in the table beats the use built on imagination). The surprise monitor catches 2/8 at 2/42 failures. The hierarchical planner [14] shows no crossover. The conformal surprise band, the re-ranking of chunk candidates, the action-conditioned risk head [11] and the calibrated surprise reflex each miss their bar by the margin in Table 1.

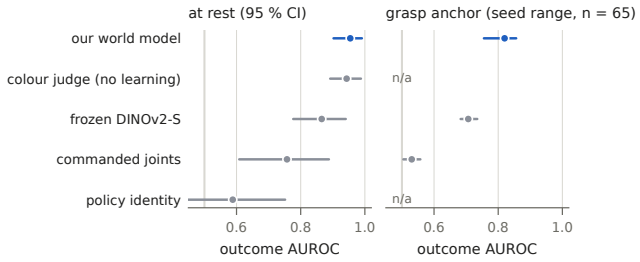


Figure 1: Table 2 as a dot plot: level with the colour judge at rest, ahead of both controls at the anchor. The right panel shows the seed range, not a CI.

4 What the encoder knows

The test here is whether the encoder carries anything that a frozen encoder or the joints do not. Table 2 sets our model beside the four other signals on the same 69 rollouts. Every probe is a LOSO linear probe against the author's labels. At rest the interval is a 95 % episode-bootstrap CI; at the anchor it is a bootstrap of the mean over 5 seeds of the out-of-fold AUROC (episodes resampled jointly across the seeds, 2,000 resamples, percentile).

At rest our model is level with the colour judge (the CIs overlap) and above the frozen encoder and the joints. At the grasp anchor no judge applies, and our model leads both controls. Since the pose probes hold the joints better than our encoder does but the outcome worse, the encoder carries outcome information they lack, at least on these rollouts.

The anchor column depends on how the five seeds are combined. Averaging the predictions before scoring gives 0.8377, 0.7175 and 0.5346 for our model, frozen DINOv2-S and the joints. Seed 0 alone gives 0.7554, 0.7024 and 0.5574, a smaller gap between our model and the frozen encoder. The order holds either way.

One control does beat our model. On SURVEYOR's scale check, frozen DINOv2-S (τ 0.3601) and the joints (τ 0.0265) pass where our model fails (τ 1.2488, against 0.20 reported by SURVEYOR). The same task and the same rollouts pass with other features, so the failure points at the geometry of our latent space rather than at the task.

The register's own outcome-readout row reports 1.000. That number was scored against the colour judge's labels, on 87 rollouts, so it partly measures agreement with the judge. Against the author's labels on the 69 rollouts, with the controls beside it, the same readout gives 0.9545.

Table 1: The register: every use of our world model re-tested here (with the frozen-DINOv2 and RoboJEPA rows beside it), what it takes from the model, whether its bar came before the result, and the verdict. Read the takes column against the verdict: the predictor uses fail. Bar, result and verdict are quoted from repro/register/expected/FEASIBILITY.md §2.

use	re-tests	takes	bar first	bar	result	verdict	code
open-loop latent prediction	[1]	predictor	yes	beats copy-last; ratios ≥ 1.3	open loop K = 30: 0.31 vs copy-last 0.89; timing shifts 1.99 / 1.58; rank 24–30	**pass**	G0
imagined- vs current-latent outcome probe (first run)	[9]	both	yes	imagined beats current	imagined 0.72, current 0.68; policy identity 0.76	not proven (policy confound)	C1
action-chunk ranking	[1]	predictor	yes	top-1 ≥ 0.8	top-1 best 0.60; action-only 0.68–0.90, regret ≈ 0 (52)	**fail**	G1b
arm-pose linear probe	[1, 8]	encoder	yes	> 0.57	arm-pose R^2 ours 0.57, **DINOv2-S 0.92**	DINOv2-S pass	G1
chunk ranking with a frozen-DINOv2 world model	[8]	predictor	yes	≥ 0.8	DINOv2-S top-1 0.55–0.61	**fail**	G1-W2
progress monotonicity and done detection	this project	encoder	yes	$\rho \leq -0.8$; AUROC ≥ 0.95	monotonicity -0.32 to -0.50; done AUROC ours 0.97 (easy contrast); 32 % false stalls	T1 **fail**, T2 sanity	G6a
outcome readout at rest; imagined-latent anticipation	[9]	both	yes	≥ 0.90; $+0.10$	readout at rest AUROC ours 1.000, DINOv2-S 0.976; anticipation $+0.179$ over policy identity	**pass** (retrospective)	G10
two-level hierarchical progress	[10]	encoder	same commit	≤ -0.80; ≤ 10 %	H-JEPA level 2 ρ -0.727; false stalls 52 %	**fail**	G6b
surprise monitor	this project	predictor	yes	$\geq 5/8$ at $\leq 1/20$	L2 surprise 2/8 at 2/42 (AUROC 0.86); drift 0/8; live reflex 42/42 successes flagged	**fail**	W1-A
latent stage tracker	this project	encoder	yes	≤ 2 %, ≤ 10 %	22.1 % premature, 31.1 % missed	**fail**	W1-B
calibrated surprise reflex	this project	predictor	yes	recall ≥ 50 %	3/25 failures at 1/44	**fail**	G11-B-cal
imagined-latent grasp probe	[9]	both	yes	≥ 0.85, $\geq +0.05$	imagined 0.70 < current 0.82	**fail**	W1-C1
conformal surprise band	this project	predictor	yes	$\leq 1/20$, ≥ 50 %	successes 22.7 % flagged, failures 46 %	**fail**	W1-C2
re-ranking chunk candidates	[1]	predictor	yes	≤ 0.10	regret 0.2631 vs ours 0.2630, action-only -0.0075	**fail**	W1-R
action-conditioned risk head	[11]	both	yes	$\geq 5/8$ at $\leq 1/20$	first version 0.93 withdrawn (time leak); second version 3/8 at 5/42, AUROC 0.79 vs shuffled 0.71	**fail**	REAP
inverse dynamics from a latent pair	this project	encoder	yes	random-init < half ours	latent pair $+0.748$ over one frame, but random-init ours 1.254 vs ours 1.180; beyond state $+0.007$	**not proven**	§30
linear identifiability of the latent	[12]	encoder	yes	$R^2 < 0.80$; ≥ 0.20	ours mean R^2 0.565, anisotropy 0.639; DINOv2-S 0.867 / 0.301	B1–B3 met	LeJEPA
latent-distance scale for plan consumption	[13]	encoder	yes	$\tau \leq 0.40$; $\geq 5/8$ at $\leq 1/20$	τ 1.249 (paper 0.20); monitor 3/8 at 6/42	**fail**	SURVEYOR
hierarchical vs flat prediction horizon	[14]	predictor	yes	crossover in $[1, 2]$ s	no crossover: high level 0.76 vs ours 0.56 at 2 s	H1, H2 **fail**	HWM
on-device latency of RoboJEPA / V-JEPA 2.1	[2, 3]	both	yes	≤ 200 ms; ≥ 15 %	encoder 526 ms (2 views); $\leq$ copy-last + 12 %	**no-go**	RoboJEPA

Table 2: Outcome AUROC of each signal on the same 69 rollouts (exploratory: controls chosen after the data). Look at the anchor column, where no judge applies.

feature	at rest	grasp anchor (65 scored, 21 fail)
our world model	0.9545 [0.9023, 0.9908]	0.8199 [0.7141, 0.9041]
colour judge (no learning)	0.9432 [0.8919, 0.9878]	—
frozen DINOv2-S	0.8655 [0.7767, 0.9405]	0.7067 [0.5816, 0.8269]
commanded joints	0.7573 [0.6084, 0.8880]	0.5305 [0.3709, 0.6855]
policy identity	0.5882 [0.4084, 0.7515]	—

5 Where the predictor fails

ContactGuard [9] finds that probes on the imagined latent beat probes on the current one for grasp tasks, so we expected the imagined latent to help at our grasp anchor. The bar asked it for ≥ 0.85, $\geq +0.05$ over the current latent. With the same encoder and the same anchor, the current latent reads the outcome at 0.82 and the imagined one at 0.70. This pre-registered row fails: on these rollouts, imagining forward loses outcome information the encoder had.

Our model's inverse-dynamics term is no better than that of its own untrained network. On the distillation probe, pooled over the 25 teleoperation validation episodes, the trained model scores 0.7476 and its random initialisation 0.7974, a difference of -0.0499 with a 95 % paired episode-bootstrap CI of $[-0.7403, +0.6559]$ (`gap5_pooled_rerun.json`). The term itself is fragile:

Table 3: VLA-JEPA world-model loss at 5 anchors with true, shuffled (Δ shuffled) and zeroed action tokens (exploratory spot check). The loss barely moves although the shuffled tokens differ by 21 % or more.

anchor (episode / frame)	true loss	Δ shuffled (%)	token change (%)	Δ zeroed (%)
8 / 402	1.3031	+0.014	21.1	+2.22
10 / 227	1.4641	+0.088	27.6	+0.50
23 / 327	1.4215	+0.036	21.1	+0.87
59 / 302	1.4637	+0.030	25.5	+1.28
65 / 302	1.5236	−0.070	22.0	−0.38

the same untrained network scored 1.4383 on one machine and 0.7974 and 0.8042 on two others (it is sensitive to the BLAS library and to ridge conditioning).

A low score can also come from a bug, so we checked the surprise monitor's failure-class AUROC (0.775, below its sanity bar) for one in loading or scoring. The same scorer reproduces the open-loop corruption ratios at K = 30: cross-episode 3.428 against 3.341 in the original run, constant-first 2.558 against 2.543 (`M1_diag_g0_repro.json`). The low AUROC is a real value.

The predictor of VLA-JEPA barely uses its actions. At 5 anchors (Table 3, from `M3_diag_action_shuffle.json`), action tokens taken from another episode differ from the true ones by 21–28 % in relative L2 but move the loss by only about 0.05 %. Zeroing the tokens moves it by about 1.05 % on average, and the forward pass is deterministic. Its failure-class AUROC of 0.501, below the same sanity bar, is therefore real, and we leave its table unread under the frozen bar.

A reader may object that one pass in the register is predictive. In the anticipation step of the outcome-readout gate, the imagined latent beat policy identity by +0.179. That bar is retrospective, though, and the two reruns of the same idea with the bar committed first, the grasp probe and the calibrated surprise reflex, both fail.

6 Related work

LeWorldModel [1] trains the recipe we use end to end from pixels and plans with it in Push-T and in simulation. Its two planning uses, chunk ranking and reranking, fail on our arm. LeJEPA's identifiability analysis [12] predicts that SIGReg latents identify the state up to anisotropy; our pose probes support that only in part. DINO-WM [8] plans on frozen DINOv2 [7] patches, which is why our frozen control is built that way. V-JEPA 2.1 [3] and RoboJEPA [2] are too slow on our on-board computer; timing them is the one register row that is not a use of our model. ContactGuard [9] is the result we contradict most directly: it finds imagined-latent probes better than current-latent ones on grasp tasks, and we find the reverse. The re-tests of ReAP's action-conditioned risk [11], the hierarchical planners of HWM [14] and H-JEPA [10] and SURVEYOR's latent-distance scale [13] all miss their bars here. Two papers agree with us from the control side. SaiVLA-JEPA [15] reports that a training-only future-latent head adds +0.03 points to control on 10,030 paired episodes, and HIJP [16] finds LeWM below goal-conditioned behaviour cloning in simulation.

7 Limitations

The claim most likely to move is the split between encoder and predictor, because it rests on probes chosen after the data and on 69 rollouts labelled by the author. Fresh rollouts scored under a bar committed first would test it. They need a day on the arm, which is not scheduled yet. The second risk is scale: everything comes from one model at one data scale, and a second point at 653 episodes is costed and queued. Three controls were not run (the surprise monitor and the imagined arm of the grasp probe would each need a predictor trained on frozen DINOv2-S features or on the joints, and progress monotonicity with the controls would need a reimplementation). The outcome-readout verdict is retrospective, and the two-level hierarchical progress use has its bar in the same commit as its result. Ordered by commit alone, without the audit, the register would read 13 bar-first and 6 same-commit rows.

8 Conclusion

Before building on the predictor of a world model this size, probe its encoder and keep the commanded actions as a baseline. On our 182 demonstrations the current latent read the grasp outcome better than the imagined one, the commanded actions alone ranked chunks better than the predictor, and every forecasting use missed its bar. A predictor may earn its place at a larger data scale, which is the next point we will measure.

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